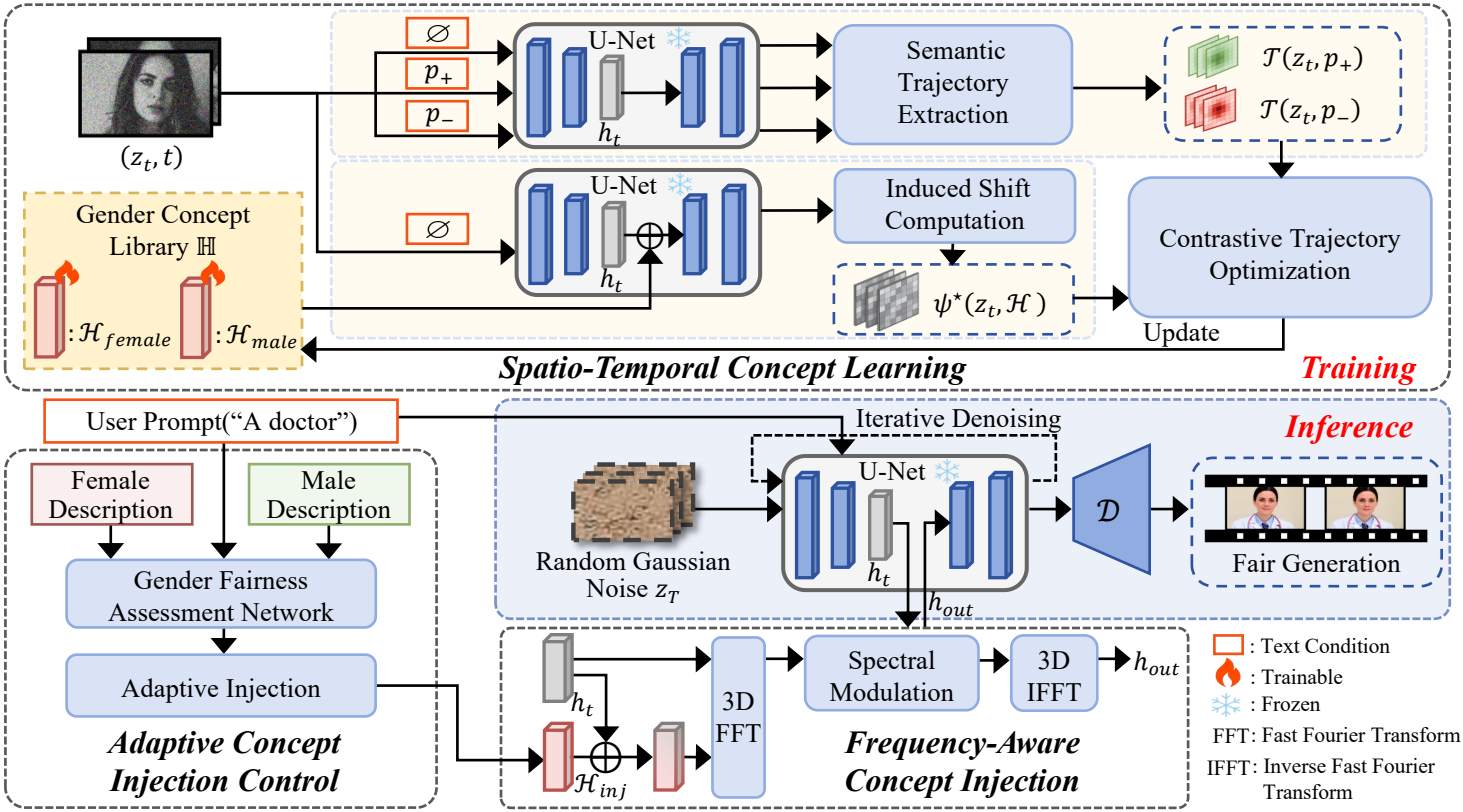
(z_t, t)
$\varnothing$
p_+
p_-
U-Net
h_t
Semantic Trajectory Extraction
$\mathcal{T}(z_t, p_+)$
$\mathcal{T}(z_t, p_-)$
Gender Concept Library $\mathbb{H}$
$: \mathcal{H}_{female}$
$: \mathcal{H}_{male}$
Induced Shift Computation
$\psi^\star(z_t, \mathcal{H})$
Contrastive Trajectory Optimization
Update
Spatio-Temporal Concept Learning
Training
User Prompt("A doctor")
Female Description
Male Description
Gender Fairness Assessment Network
Adaptive Injection
Adaptive Concept Injection Control
Iterative Denoising
Inference
Random Gaussian Noise z_T
h_{out}
$\mathcal{D}$
Fair Generation
$\mathcal{H}_{inj}$
3D FFT
Spectral Modulation
3D IFFT
Frequency-Aware Concept Injection
: Text Condition
: Trainable
: Frozen
FFT: Fast Fourier Transform
IFFT: Inverse Fast Fourier Transform